Lattice Path Sum and the Continuum Klein–Gordon Propagator

Vahid Forghanipour

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1 Introduction and Random Walk Formulation

Consider a four-dimensional hypercubic lattice with lattice spacing Δ. A lattice site is labelled by an integer vector

$$n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4, \qquad x = n\Delta.$$

A nearest-neighbor step changes one component of n by ± 1. Consequently, every lattice site has

$$2d = 8$$

possible outgoing links.

For a path consisting of L links, assign the Boltzmann weight

$$\exp(-m_0 L),$$

where m_0 is the dimensionless bare mass associated with one lattice step. The lattice path sum from the origin to n is therefore

$$G_\Delta(n) = \sum_{L=0}^{\infty} e^{-m_0 L} \sum_{\substack{\text{nearest-neighbor paths} \\ 0 \to n,\ \text{length } L}} 1.$$

Paths may retrace their steps and may traverse any link any number of times. This is precisely the unrestricted lattice random-walk formulation. The use of the Euclidean path-length action

$$S[\text{path}] = m_0 L[\text{path}]$$

is the lattice analogue of the relativistic-particle construction in Section 6, Lecture 7 of *Zamolodchikov's lecture notes*, where the continuum action is $A[\text{path}] = m_0 L[\text{path}]$ (pp. 53–56).

2 Fourier Representation of the Lattice Propagator

Let a single lattice step be represented by a vector

$$\epsilon \in \{\pm e_1, \pm e_2, \pm e_3, \pm e_4\},$$

where e_μ is the unit vector in the μth lattice direction. For a path with L steps,

$$n = \sum_{j=1}^{L} \epsilon_j.$$

Using the Fourier representation of the Kronecker delta,

$$\delta_{n,0} = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} e^{ik\cdot n},$$

the constraint that the total displacement equals n can be imposed as

$$\delta_{n,\sum_{j=1}^{L}\epsilon_j} = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} \exp\left[ik \cdot \left(n - \sum_{j=1}^{L} \epsilon_j\right)\right].$$

Summing independently over the possible steps gives

$$\sum_{\epsilon} e^{-ik\cdot\epsilon} = \sum_{\mu=1}^{4} \left(e^{-ik_\mu} + e^{ik_\mu}\right) = 2\sum_{\mu=1}^{4} \cos k_\mu.$$

Hence

$$\begin{aligned} G_\Delta(n) &= \sum_{L=0}^{\infty} e^{-m_0 L} \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot n} \left(2\sum_{\mu=1}^{4} \cos k_\mu\right)^L \\ &= \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} \frac{e^{-ik\cdot n}}{1 - 2e^{-m_0}\sum_{\mu=1}^{4} \cos k_\mu}. \end{aligned} \tag{1}$$

Thus the exact lattice propagator is

$$\boxed{G_\Delta(n) = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} \frac{e^{-ik\cdot n}}{1 - 2e^{-m_0}\displaystyle\sum_{\mu=1}^{4} \cos k_\mu}}$$

The sign in the Fourier exponential is conventional and can be reversed by replacing $k \mapsto -k$.

3 Divergence, Criticality, and Bare Mass Tuning

The geometric series over path lengths converges whenever the largest possible magnitude of the one-step eigenvalue satisfies

$$e^{-m_0} \max_{k\in[-\pi,\pi]^4} \left|2\sum_{\mu=1}^{4} \cos k_\mu\right| < 1.$$

Since

$$\max_{k} \left|2\sum_{\mu=1}^{4} \cos k_\mu\right| = 8,$$

the convergence condition is

$$8e^{-m_0} < 1.$$

Therefore,

$$m_0 > \log 8,$$

and the critical bare mass is

$$m_{0,c} = \log 8.$$

At $m_0 = m_{0,c}$, the denominator vanishes at $k = 0$:

$$1 - 2e^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu \Big|_{k=0} = 1 - 8e^{-m_0} = 0.$$

The zero-momentum mode then produces an infrared divergence.

For $m_0 < m_{0,c}$, the entropy of long paths overcomes their Boltzmann suppression. The number of paths grows asymptotically as 8^L, whereas their weight is $e^{-m_0 L}$. Thus the large-L behavior is controlled by

$$(8e^{-m_0})^L,$$

which grows exponentially when $m_0 < \log 8$. The unrestricted path sum therefore diverges.

For $m_0 > m_{0,c}$, long paths are exponentially suppressed and the sum is dominated by relatively short paths. In the limit $m_0 \to +\infty$, only the shortest paths survive. This is the frozen, or noncritical, limit. However, simply taking a fixed $m_0 > m_{0,c}$ while $\Delta \to 0$ does not produce a nontrivial continuum propagator. The bare mass must be tuned toward criticality.

4 Continuum Limit and Physical Klein–Gordon Propagator

Introduce physical coordinates and momenta by

$$x = n\Delta, \qquad p = \frac{k}{\Delta}, \qquad k = \Delta p.$$

Then

$$d^4k = \Delta^4 d^4p, \qquad e^{-ik\cdot n} = e^{-ip\cdot x}.$$

For small k,

$$\cos k_\mu = 1 - \frac{k_\mu^2}{2} + O(k_\mu^4),$$

and consequently

$$2\sum_{\mu=1}^{4} \cos k_\mu = 8 - k^2 + O(k^4), \qquad k^2 = \sum_{\mu=1}^{4} k_\mu^2.$$

Write the bare mass in the form

$$e^{-m_0(\Delta)} = \frac{1}{8} \exp\left[-\frac{m^2\Delta^2}{8}\right].$$

Equivalently,

$$m_0(\Delta) = \log 8 + \frac{m^2\Delta^2}{8}.$$

Then

$$\begin{aligned} 1 - 2e^{-m_0(\Delta)} \sum_{\mu=1}^{4} \cos k_\mu &= 1 - \exp\left[-\frac{m^2\Delta^2}{8}\right]\left(1 - \frac{k^2}{8} + O(k^4)\right) \\ &= \frac{k^2 + m^2\Delta^2}{8} + O(k^4, \Delta^4, k^2\Delta^2). \end{aligned} \tag{2}$$

Since $k = \Delta p$,

$$1 - 2e^{-m_0(\Delta)} \sum_{\mu=1}^{4} \cos(\Delta p_\mu) = \frac{\Delta^2}{8}\left(p^2 + m^2\right) + O(\Delta^4).$$

Therefore,

$$\begin{aligned} G_\Delta(n) &= \Delta^4 \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip\cdot x}}{\frac{\Delta^2}{8}(p^2 + m^2)} + O(1) \\ &= 8\Delta^2 \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip\cdot x}}{p^2 + m^2} + O(1). \end{aligned} \tag{3}$$

The dimensionful continuum propagator is therefore obtained with the normalization

$$G_{\text{cont}}(x) = \lim_{\Delta\to 0} \frac{\Delta^2}{8} G_\Delta\left(\frac{x}{\Delta}\right).$$

Thus,

$$\boxed{m_0(\Delta) = \log 8 + \frac{m^2\Delta^2}{8} + O(\Delta^4)}$$

and

$$\boxed{G_{\text{cont}}(x) = \lim_{\Delta\to 0} \frac{\Delta^2}{8} G_\Delta\left(\frac{x}{\Delta}\right) = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip\cdot x}}{p^2+m^2}}$$

The additive divergence $\log 8$ is the entropy contribution from the eight possible directions at every step. The finite $O(\Delta^2)$ part determines the physical mass. This is the lattice random-walk version of bare-mass renormalization discussed in *Zamolodchikov's lecture notes*, pp. 54–56.

The precise coefficient in the relation between m_0 and m depends on the microscopic step distribution and normalization. For the nearest-neighbor hypercubic walk used here, the coefficient is $1/8$. Other discretizations can have different cutoff relations while yielding the same continuum Klein–Gordon propagator after the appropriate tuning.

5 Random Walk Interpretation: $L_{\text{typ}} \sim \Delta^{-2}$

For an unbiased random walk, the displacement after L independent steps has zero mean:

$$\langle n_\mu \rangle = 0.$$

Each step contributes a variance of order one, so

$$\langle n^2 \rangle = \left\langle \sum_{\mu=1}^{4} n_\mu^2 \right\rangle \sim L.$$

Consequently,

$$|n|_{\text{typ}} \sim \sqrt{L}.$$

The physical displacement is

$$|x|_{\text{typ}} = \Delta |n|_{\text{typ}} \sim \Delta\sqrt{L}.$$

To obtain a finite nonzero physical displacement as $\Delta \to 0$, one must require

$$\Delta\sqrt{L} \sim 1.$$

Therefore,

$$\boxed{L_{\text{typ}} \sim \Delta^{-2}}$$

This explains why the naive continuum limit with fixed physical length

$$L\Delta = \text{constant}$$

is not the correct limit for the paths that contribute to the continuum propagator. Such paths have

$$L \sim \Delta^{-1},$$

and hence

$$|x|_{\text{typ}} \sim \Delta\sqrt{\Delta^{-1}} = \sqrt{\Delta} \longrightarrow 0.$$

They collapse onto the origin rather than producing finite propagation.

The relevant finite scaling variable is instead

$$s \sim L\Delta^2.$$

This is the proper-time or Brownian parameter. It remains finite when

$$\Delta \to 0, \qquad L \to \infty, \qquad L\Delta^2 = \text{constant}.$$

In the continuum, the contributing paths therefore have divergent microscopic length,

$$L\Delta \sim \Delta^{-1} \longrightarrow \infty,$$

but finite macroscopic displacement. They are highly crumpled and have the characteristic Brownian fractal dimension two. This agrees with the proper-time scaling described in *Zamolodchikov's lecture notes*, pp. 54–56.

Using the Schwinger parameter representation,

$$\frac{1}{p^2+m^2} = \int_0^\infty ds\, e^{-s(p^2+m^2)},$$

the continuum propagator can also be written as

$$G_{\text{cont}}(x) = \int_0^\infty ds\, e^{-m^2 s} \int \frac{d^4p}{(2\pi)^4} e^{-sp^2} e^{-ip\cdot x}.$$

The factor $e^{-m^2 s}$ suppresses large proper times, while the Gaussian factor e^{-sp^2} encodes Brownian spreading.

6 Final Conclusion and Summary of Equations

The unrestricted nearest-neighbor path sum on a four-dimensional hypercubic lattice is a geometric series in the one-step Fourier eigenvalue

$$2\sum_{\mu=1}^{4} \cos k_\mu.$$

Its exact momentum-space form is

$$\boxed{G_\Delta(n) = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} \frac{e^{-ik\cdot n}}{1 - 2e^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu}}$$

The entropy of the walk gives the critical threshold

$$m_{0,c} = \log 8.$$

The path sum converges for

$$m_0 > \log 8,$$

diverges for

$$m_0 < \log 8,$$

and becomes critical at

$$m_0 = \log 8.$$

To obtain a finite continuum limit with physical mass m, the bare mass must be tuned toward criticality according to

$$\boxed{m_0(\Delta) = \log 8 + \frac{m^2\Delta^2}{8} + O(\Delta^4)}$$

The lattice and continuum propagators are related by

$$G_{\text{cont}}(x) = \lim_{\Delta\to 0} \frac{\Delta^2}{8} G_\Delta\left(\frac{x}{\Delta}\right),$$

which gives

$$\boxed{G_{\text{cont}}(x) = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip\cdot x}}{p^2 + m^2}}$$

Finally, finite physical propagation requires Brownian scaling:

$$\boxed{L_{\text{typ}} \sim \Delta^{-2}}$$

rather than fixed physical path length $L\Delta$. The continuum paths have infinitely many microscopic steps and divergent microscopic length, while their root-mean-square physical displacement remains finite.

AI use declaration

Model(s) used: GPT-6 Astra

Revisions/corrections made to model output, and clarifications added:GPT-6 Astra

Comparisons/checks against the lecture notes:
Imaginary-time formalism and euclidean paths chapter5 p.38 also chapter 6 pp. 51-52, ,concept of propagator and discretized path chapter 5 pp.40-42, concept of bare mass and its difference with actual mass chapter 6 p.52, discretized path sum and end point constraint p.52 and Fourier representation as an integral over four-momentum, bare mass tuning chapter 6 p.54

Calculation steps, limiting cases, or consistency checks verified by hand:
Equation of propagator Eq.1 p.2, divergence limit of bare mass p.3, bare mass tunnig equation p.4

Calculation steps, limiting cases, or consistency checks subjected to automated verification (via Mathematica, python, LLMs):-